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Impact of Literacy Interventions on Reading Skills in Low- and Middle-Income Countries: A Meta-Analysis

Young-Suk G. Kim[a], Hansol Lee[b], Stephanie S. Zuilkowski[c]

[a] *University of California, Irvine*
[b] *Korea Military Academy*
[c] *Florida State University*

Correspondence concerning this article should be addressed to Young-Suk G. Kim, School of Education, University of California, Irvine, 92697. Electronic mail may be sent to youngsk7@uci.edu.

Abstract

Reading skills are foundational for daily lives, academic achievement, and careers. In this study, we systematically reviewed literacy interventions in low- and middle-income countries, and estimated their effects on children's reading skills using a meta-analytic approach. A total of 67 studies (N = 213,464) from 32 countries found in various databases (e.g., PsycINFO, ERIC) and sources (e.g., United States Agency for International Development) met our inclusion criteria. The results revealed an overall effect of .30 across various literacy outcomes. Effects varied for different outcomes, such that largest effects were found in emergent literacy skills (e.g., .40) and the smallest effects in reading comprehension (.25) and oral language skills (.20). Effects also varied as a function of other features such as teacher training support.

Reading is one of the core foundational skills for academic achievement and careers in the modern information-driven society. As such, development of reading skills has critical implications for individual children. For example, students who enter the upper elementary grades as poor readers continue to struggle throughout their schooling (Francis, Shaywitz, Stuebing, Shaywitz, & Fletcher, 1996) and are at high risk for academic failure, drop out, unemployment, overall low income, criminal activity, and even poor health (Christenson & Thurlow, 2004; Daniel et al., 2006; Hernandez, 2011). Beyond growth and advancement for an individual, reading skills also have economic consequences at the societal level. For instance, a country's performance on international assessments on reading and content areas (e.g., math) accounts for more than 70% of the variation in economic growth (Hanushek & Woesman, 2012).

Unfortunately, however, many children, particularly those in low- and middle-income countries (LMICs—see below for operational definition following the World Bank Atlas method), do not develop foundational reading skills (e.g., decoding words) even after years of instruction (e.g., UNESCO, 2012; Uwezo, 2012). According to recent United Nations (UN) data, 9 of 10 children (i.e., 202 million children) in Sub-Saharan Africa do not have basic reading skills. A similarly dire situation is also found in Central and Southern Asia, where approximately 81% of children (or 241 million children) do not read with minimum proficiency (UNESCO, 2017b). In response to this reality, stakeholders in LMICs have paid unprecedented attention to reading development and instruction in primary grades in the last decade along with various initiatives such as the Sustainable Development Goals and the Education for All initiative by UN agencies, donors, and nongovernmental organizations.

The goal of this study was to review the effects of reading interventions in LMICs on reading and language outcomes, using a meta-analytic approach. Considerable research in the past 4 decades has revealed that reading involves development and coordination of multiple language, cognitive, and print-related skills (e.g., Cromley & Azevedo, 2007; Kim, 2017; Perfetti & Stafura, 2014; Vellutino, Tunmer, Jaccard, & Chen, 2007). Notably, the National Reading Panel (National Institute of Child Health and Human Development, 2000), after an extensive review of evidence available until the late 1990s, summarized key areas of reading instruction as follows: phonological awareness, phonics, reading fluency, vocabulary, and reading comprehension. Phonological awareness and phonics are necessary for lexical level reading skill

(or decoding), because in languages with alphabetic writing systems, graphemes represent sounds and therefore, deciphering the code requires an understanding of phonological structure (e.g., syllable, phoneme; phonological awareness) and knowledge of correspondences between orthographic symbols and sounds (Adams, 1990; National Early Literacy Panel, 2008; National Institute of Child Health and Human Development, 2000). Reading fluency, one's ability to read words accurately and rapidly with appropriate expression in connected texts, is also necessary for reading comprehension as it allows cognitive resources such as attention and working memory to be available for comprehension processes (Fuchs, Fuchs, Hosp, & Jenkins, 2001; Kim, Petscher, Schatschneider, & Foorman, 2010; Kim & Wagner, 2015; Jenkins, Fuchs, van den Broek, Espin, & Deno, 2003; Riedel, 2007; Roehrig, Petscher, Nettles, Hudson, & Torgesen, 2008). Oral language skills such as vocabulary are also crucial for comprehending decoded text (Elleman, Lindo, Morphy, & Compton, 2009; National Institute of Child Health and Human Development, 2000). Furthermore, although not included in the National Reading Panel Report due to lack of evidence at the time, recent studies have revealed that oral language skills beyond vocabulary such as grammatical knowledge and listening comprehension as well as higher order cognitive skills (e.g., inference, perspective taking) are necessary for reading comprehension (e.g., Alonzo, Yeomans-Maldonado, Murphy, & Bevens, 2016; Florit, Roch, Altoè, & Levorato, 2009; Florit, Roch, & Levorato, 2014; Kendeou, Bohn-Gettler, White, & van den Broek, 2008; Kim, 2017).

Theory as well as evidence from developed countries indicates that explicit and systematic instruction on the appropriate components noted above improves children's reading and reading-related skills (e.g., oral language; National Institute of Child Health and Human Development, 2000). However, findings from developed countries cannot be assumed to readily generalize to LMICs as these countries face numerous challenges at multiple levels. At the macro level, ministries of education often lack sufficient resources to support schools, train teachers, and conduct data collection and management. On the ground, large class sizes are common; the primary level student–teacher ratios in Chad, Malawi, and Rwanda are 62:1, 70:1, and 58:1, respectively (UNESCO, 2017a). The ratios of students to trained and qualified teachers are often far larger. In Madagascar there is one trained primary teacher for every 273 pupils, for example (UNESCO, 2017a). Schools are often under-resourced with lack of electricity, furniture, books, paper and pencils, and even buildings (Lee & Zuilkowski, 2015). Many teachers are poorly trained (Akyeampong, Lussier, Pryor, & Westbrook, 2013) and earn low salaries, which can erode their job satisfaction (Gamero Burón & Lassibille, 2016). In many places, teacher and student absenteeism rates are high, reducing children's opportunity to learn (Benavot & Gad, 2004; Dubeck, Jukes, & Okello, 2012). The adult literacy rate is 61% in low-income countries (UNESCO, 2017a), which has implications for children's learning in the home and modeling of literacy behaviors. Furthermore, many countries are experiencing conflict, unrest, and crisis, with severe implications for education (Collins & Messaoud-Galusi, 2012; Gove & Cvelich, 2011; Rugh, 2012; UNICEF and UNESCO Institute of Statistics, 2014; Verger, Novella, & Altinyelken, 2012). All these factors, typically not found in high-income countries, present serious challenges for implementing reading interventions and improving children's learning.

A recent systematic narrative review of reading interventions in LMICs revealed that many studies found positive effects of reading interventions, although some studies also showed null effects (a narrative review does not involve statistical analysis of effect sizes [ESs]; Kim, Boyle, Zuilkowski, & Nakamura, 2016). Effects tended to be particularly consistent and large for lower level skills (or emergent literacy skills), such as phonological awareness and letter knowledge, whereas inconsistent and relatively small effects were found in higher order skills such as reading comprehension (Kim et al., 2016). Similarly, Graham and Kelly (2018) recently reviewed 18 studies in developing countries and found a large mean ES of .63 in letter sound knowledge compared to a smaller ES, .34, in reading comprehension. McEwan (2015) reviewed randomized controlled trials (RCTs) in developing countries that focused on health and nutrition interventions (e.g., deworming or providing school lunch) and instructional interventions. Overall, he found that the largest average ESs were for programs that included instructional materials (.08), teacher training (.12), and smaller classes or ability grouping (.12), whereas health related interventions yielded null average effects on children's learning. Finally, Conn (2017) evaluated the impact of educational interventions such as health treatment, provision of school supplies, the use of teacher incentives, and schoolbased management programs on student learning outcomes in sub-Saharan African contexts. She found that programs that targeted changing classroom instruction had the most consistent and positive average ES, .30.

These previous reviews and meta-analyses suggest that ESs vary as a function of several factors. For instance, one important factor in intervention is the language in which reading is taught. In many LMICs, children learn to read in secondary language (s) (L2 hereafter) rather than the child's first language or mother tongue (L1). Reading acquisition, like any learning, requires comprehensible input and therefore, is easier in a child's most familiar language. Although a body of evidence indicates the effect of L1 literacy skills on L2 literacy acquisition (Vaughn et al., 2006; Wawire & Kim, 2018; see Ball, 2011 for a review), literacy instruction in many LMIC contexts is provided in only L2. This is the result of myriad political and cultural factors, often the legacies of colonial education systems, as well as practical reasons, such as the availability of teachers trained to teach in L1 and sufficient teaching materials. Given these complexities, implementing literacy instruction in L1 can be challenging in many LMICs. Then, a critical question is whether effects of literacy interventions differ when instruction is in L1 versus L2.

Interventions can vary in several other aspects as well. In some interventions, teachers are provided with a one-off teacher training workshop, whereas in others, ongoing support is provided, including regular classroom visits and feedback and booster training sessions. Although many previous interventions included one-off training workshops, some included ongoing sustained support such as refresher training sessions and in-school teacher coaching and found it to be effective in improving children's learning outcomes (Education Development Center, 2014a, 2014b; Piper & Korda, 2011; Pouezevara, Costello, & Banda, 2013). Interventions also vary in terms of involvement of the community in student learning. The vast majority of interventions target instruction in the school context only, whereas some incorporate community involvement to encourage literacy-related activities in the children's homes and communities (e.g., Banerjee et al., 2010; Dowd & Friedlander, 2016; Friedlander & Goldenberg, 2016). According to the Ecological Systems Theory, interactions in multiple layers of environment shape human development (Bronfenbrenner, 1979), and therefore, children's daily interactions in their homes and communities beyond the school setting influence their learning, including literacy skills (Bracken & Fischel, 2008; Senechal, LeFevre, Thomas, & Daley, 1998). Interventions also vary in other aspects such as intervention duration and target subjects. Some interventions last less than a single academic year (e.g., Jere-Folotiya et al., 2014 for 2 weeks on the effect of information and communications technology [ICT] on initial literacy skills; Friedland, Gilman, Johnson, & Demeke, 2017 for 30 days on reading fluency; Abrami, Wade, Lysenko, Marsh, & Gioko, 2016 for 13 weeks on the effect of multimedia software on literacy), whereas others are implemented across multiple years (e.g., Banerjee, Cole, Duflo, & Linden, 2007 for 1 year; Jonason, Guajardo, Nath, & Hossain, 2014 and RTI International, 2015a for 2 years of implementation; Courtney & Gravelle, 2014 for a period of 5 years). Some interventions target only reading (e.g., Abrami et al., 2016; Courtney & Gravelle, 2014; Friedland et al., 2017; Jere-Folotiya et al., 2014; Jonason et al., 2014), whereas others target both reading and math (e.g., Banerjee et al., 2007; RTI International, 2015a).

Beyond these intervention characteristics, results may be moderated as a function of contextual features such as urbanicity (e.g., Badiable, Guajardo, Fermin, & Robis, 2013; Dunlop, 2015 for urban areas; Courtney & Gravelle, 2014; Friedland et al., 2017; RTI International, 2015a for rural areas) and region (e.g., Abrami et al., 2016; Friedland et al., 2017; Jere-Folotiya et al., 2014; RTI International, 2015a for Sub-Saharan Africa; Courtney & Gravelle, 2014; Dunlop, 2015 for South East Asia & Pacific; Banerjee et al., 2007; Jonason et al., 2014 for South Asia). Although patterns may vary across contexts, rural areas in many LMIC contexts struggle with school quality issues including difficulty attracting qualified teachers (Mulkeen & Chen, 2008). In Thailand, for example, school factors including qualified teachers have been found to account for nearly half of the urban–rural achievement gap (Lounkaew, 2013). Regional differentials may be due to differences is urbanicity, or related to L1 language patterns, socioeconomic disparities, or other factors. These issues lead to the question of whether intervention effects vary as a function of these characteristics.

In this study, we conducted a meta-analysis of reading interventions in LMICs to obtain accurate estimates of their effects and potential moderators, such as intervention features and contextual factors (e.g., urbanicity, region). The following were two guiding research questions: (a) What is the impact of multicomponent literacy intervention in LMICs on language and reading outcomes?; (b) do the effects vary as a function of intervention features such as target language (L1

vs. L2), the nature of professional development (PD; presence or absence of ongoing support for teachers), intervention domain (reading only or include reading and math), inclusion of ICT, community involvement, and intervention duration as well as contextual factors such as urbanicity (urban vs. rural) and region (e.g., Africa, Central and South America)?

When addressing these questions, we examined the impact of reading interventions on the overall average effect as well as the component skills individually. Specifically, we report effects on the following reading and reading-related outcomes aligned with extant literature: (a) emergent literacy skills, (b) word reading, (c) reading fluency, (d) reading comprehension, and (e) oral language. Emergent literacy skills included prereading skills such as phonological awareness, letter knowledge, and print awareness (concepts of print). Word reading included children's ability to read words, including nonwords and syllables. Reading fluency included automaticity of reading connected texts, typically measured as number of words read correctly within a specified time (e.g., 1 min). Reading comprehension included children's ability to read and comprehend texts. Oral language skills included vocabulary, syntax, and listening comprehension. These five component skills are in line with the above noted component skills of reading development based on theory and empirical evidence, and reveal the extent to which reading interventions impact these outcomes of interest. Our computation of ESs (i.e., standardized mean differences between treatment and control groups) allowed us to compute the comparable treatment effects needed for meta-analytic data analyses.

We anticipated a larger ES for lower level skills such as emergent literacy skills compared to higher level skills such as reading comprehension. We also expected that ESs might be larger for studies targeting children's first language. A larger ES was also hypothesized for studies that involved the community in improving children's literacy skills, given the consistently positive relations between language and literacy activities in the home and children's literacy acquisition (Bus, van IJzendorn, & Pellegrini, 1995; Chansa-Kabali, Serpell, & Lyytinen, 2014; Dowd & Friedlander, 2016; Dynia, Lawton, Logan, & Justice, 2014). We did not have specific hypotheses about other moderators such as inclusion of both math and reading, intervention duration, urbanicity, and region.

Method

We systemically employed the following steps to measure the effects of literacy interventions conducted in LMICs. First, we searched previous published and unpublished papers, such as journal articles, reports, and dissertations that report the impacts of literacy interventions. Second, we reviewed identified studies to confirm whether they met our inclusion criteria. Third, we coded relevant information in each study, checked for any possible publication bias or small-study effects, and computed ESs. Accuracy and reliability was verified in every step, and any discrepancy was resolved via discussions among the authors.

Literature Search and Inclusion Criteria

To be included in the present meta-analysis, studies must have been (a) literacy interventions, (b) written in English, (c) published between 2000 and 2017, (d) conducted in LMICs (i.e., low-income countries, low-middle-income countries, and upper-middle-income countries according to the World Bank's classification), (e) conducted as a RCT or quasi-experiment, and (f) with more than 20 preprimary and primary grade (up to Grade 4) children. We included studies with a clear and explicit mention and focus on literacy "instruction." For instance, a book intervention (e.g., providing more books to schools; Glewwe, Kremer, & Moulin, 2009) with assessments of literacy outcomes was excluded because it did not include instruction as part of the intervention. In contrast, book reading interventions such as dialogic reading was included in the study. If oral language was the target skill in the intervention, it had to be part of a literacy intervention.

The publication period after year 2000 reflects the fact that there were practically no reported literacy interventions prior to 2000 in this set of countries. Using the World Bank Atlas method, low-income countries are countries with a GNI per capita of $995 or less in 2017, low-middle-income countries are with a GNI per capita between $996 and $3,895, and upper-middle-income are with a GNI per capita between $3,896 and $12,055 for the current 2019 fiscal year. For more information, visit the World Bank webpage.

For the literature search, databases and websites of the literacy initiatives or funding agencies were used. Databases include PsycINFO and ERIC, and the latter (initiatives or funding agencies) include United States Agency for International Development, Save the Children, United Kingdom Department for International Development, Education Development Center, American Institutes for Research, and Pratham Education Foundation. For the search through databases, the following 10 sets of keywords were used: (a) “emergent literacy” and “intervention” or “randomized control,” (b) “phonological awareness” and “intervention” or “randomized control,” (c) “letter knowledge” and “intervention” or “randomized control,” (d) “alphabet knowledge” and “intervention” or “randomized control,” (e) “morphological awareness” and “intervention” or “randomized control,” (f) “orthographic awareness” and “intervention” or “randomized control,” (g) “phonics” and “reading” and “intervention” or “randomized control,” (h) “code based approach” and “intervention” or “randomized control,” (i) “reading fluency” or “oral reading fluency” or “text reading fluency” and “intervention” or “randomized control,” and (j) “reading comprehension” and “comprehension” or “reading for understanding” and “intervention” or “randomized control.” For websites manual searches were conducted. Along the lines, the lists of studies included in previous reviews of the similar topics were revisited (i.e., Graham & Kelly, 2018; Kim et al., 2016; McEwan, 2015).

A total of 3,861 papers were initially identified in these searches with a final set of 67 studies after the screening process according to our six inclusion criteria noted above. Figure 1 illustrates the PRISMA flowchart describing the aforementioned procedure. First, we identified 3,639 papers through database searching and an additional 200 papers through manual search. After removing duplicates, there were a total of 3,861 papers identified. Second, we screened their titles and abstracts, and found that 3,669 of them did not meet our criteria—the majority of the excluded papers came from databases via keyword searches, and they were not relevant to the main topic. Third, we assessed the full texts of the remaining 192 papers to determine their eligibility. A total of 48 papers were eligible, whereas we excluded 144 papers, including three review articles, 69 non-RCTs and nonquasi experiments, three studies with no control group, five studies irrelevant to literacy interventions, 15 studies with no literacy component outcome, three studies with students above the primary grades, two studies with small samples ($n < 20$), 2 midline reports, 22 replicate studies (i.e., studies analyzed the same data set), and 20 papers with insufficient data. We contacted authors of the 20 studies with insufficient data and acquired necessary data from authors of 19 studies. Overall, a total of 67 studies met all our criteria and they had 129 independent samples representing 661 ESs (for each measurement) from 32 countries (N = 213,464), which will be described in the following section.

Multilevel ES Calculation

A typical approach in previous meta-analyses was to compute single ESs (see Borenstein, Hedges, Higgins, & Rothstein, 2009) because calculating multiple ESs for a study may violate the assumption of independence. For example, in one study, Piper and Korda (2011) had two types of literacy treatments for the intervention, making two independent samples: (a) “full” treatment group versus control group, and (b) “light” treatment group versus control group. For each treatment, the Early Grade Reading Assessment was employed to measure children’s literacy improvement, and the test had seven subtasks: letter naming, phonemic awareness, familiar word fluency, unfamiliar word fluency, oral reading fluency, reading comprehension, and listening comprehension. However, according to the typical approach, we need to compute two ESs for each independent sample after making average ESs for different measurements.

To overcome this limitation, recent meta-analytic efforts (e.g., Lee, Warschauer, & Lee, in press; McEwan, 2015; Pratt, Turanovic, & Cullen, 2016) have used a multilevel modeling approach so that the structures of the data set can be maintained for precise estimation (see Hox, Moerbeek, & van de Schoot, 2010 for theoretical background). In this way, we were able to calculate more than one ES for each independent sample when it has more than one measurement; therefore, we computed 14 ESs for two independent samples of Piper and Korda (2011). As a result, we found that the selected 67 studies represented 129 samples and 661 ESs.

Because each study has different ways of reporting the effects of the interventions, we followed the procedures shown in Figure 2 to calculate ESs precisely and reliably. First, we checked if a study had calculated ESs regardless of the scales, and if so, we accepted the reported ESs; and when necessary, we converted them into Cohen's unbiased *d* (called Hedges' *g*; Borenstein et al., 2009; Hedges, 1981 for a review and rationale). Second, for studies without any reported ESs, we checked whether there is a baseline difference between treatment and control groups according to their pretest results. Third, if samples in a study did not have any baseline differences (e.g., RCTs), we computed ESs for posttest scores based on reported descriptive statistics. Fourth, if there were differences in baseline performances in a study, we calculated ESs with the reported inferential statistics (e.g., analysis of covariance). We also computed both posttest and gain-score (i.e., posttest scores–pretest scores) ESs if a study did not perform any statistical adjustments. For gain-score ES calculations, we used the standard deviation from the posttest control group in place of that of the gain score, which are not typically reported in a study. See Lee et al. (in press) for other possible suggestions for gain-score ES calculations. If a study did not present the necessary statistics, we contacted authors for data for the ES calculation and were able to acquire data for 19 studies (see Figure 1).

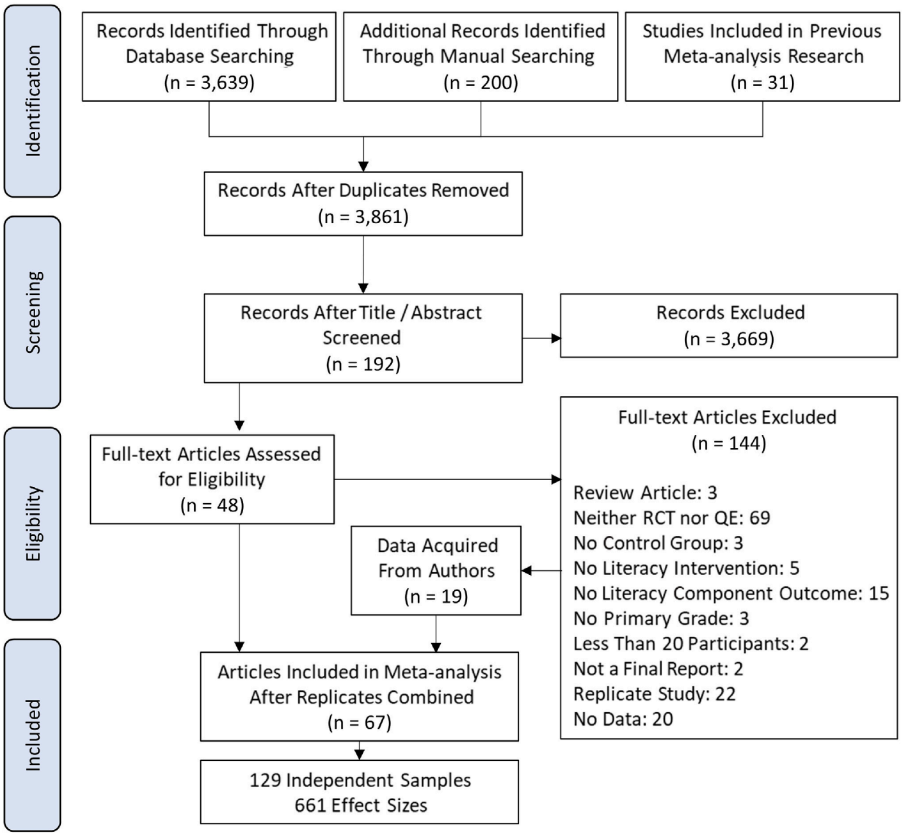


Figure 1. PRISMA flowchart (adapted from Moher, Liberati, Tetzlaff, Altman, & PRISMA Group, 2009). [Color figure can be viewed at wileyonlinelibrary.com]

Publication Bias and Small-Study Effects

Given that meta-analysis research solely relies on identifiable publications, the calculated ESs for a meta-analysis could be influenced by a phenomenon that studies with significant findings and/or large effects are likely to be reported and accepted in journals. As part of this so-called publication bias, another noticeable phenomenon is that studies with small samples might be published more frequently than large-scale studies, which probably produce more reliable results; yet, the ESs from these two different pools of studies do not necessarily resonate, and this could be a sign of small-study effects (see Egger, Smith, Schneider, & Minder, 1997 for a discussion).

To minimize these possible biases we conducted our literature search by identifying unpublished dissertations, reports, and working papers on large-scale experiments. Although our findings suggested that different publication types had different average ESs (e.g., journal articles had average ES of .46, whereas reports and working papers had average ES of .27), this difference was not related to small-study effects because the number of participants was large for the studies reported in journal articles (mean N = 2,375) as well as reports and working papers (mean N = 1,455). Furthermore, we checked whether there was any significant association between the calculated ESs and the sizes of the literacy interventions (i.e., numbers of children and schools participated in the interventions); we could not find any statistically significant correlation ($p > .05$). To take into consideration this mixed finding, the publication type and participant numbers were controlled for further analyses.

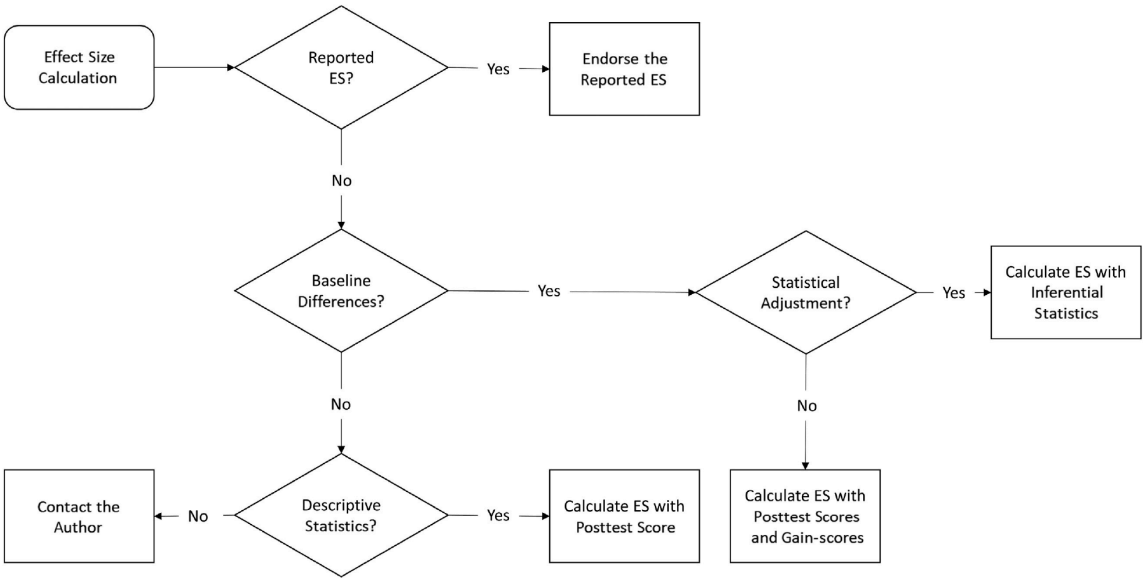


Figure 2. Effect size calculation flowchart.

We also used a funnel plot (Egger et al., 1997) as well as a trim-and-fill technique (Duval & Tweedie, 2000) to statistically detect any sign of publication bias and small-study effects. First, the key aspect in the funnel plot is the association between the ESs estimated from individual studies (plotted on the horizontal axis) and their sample sizes or standard errors (plotted on the vertical axis; see Figure 3). On the assumption that all the studies are coming from the same population (i.e., assuming an absence of bias and heterogeneity), a funnel plot should show a solid vertical line on the overall average ES and make a triangular region when adding two dotted diagonal lines representing 95% confidence limits around the average ES. If the calculated ESs—dots in Figure 3—are located within the triangular region with evenly distributed on both sides of the central solid line (i.e., symmetry), it may indicate an absence of publication bias. The funnel plot for the present meta-analysis (Figure 3a) illustrates that the group of dots has an overall symmetrical distribution, though substantial amounts of the dots are outside the triangular region.

In addition to the visual confirmation of publication bias, Egger et al. (1997) suggested a statistical method to test asymmetry in funnel plots by checking whether an intercept in a linear regression of standardized effect estimates (ES/*SE*) in precision (the inverse of standard error) is different from zero. The results of Egger's test for our calculated ESs (Figure 3b) suggested that the intercept in the linear regression is not different from zero (intercept = .78, SE = .41, $p > .05$ [95% CI: −0.04, 1.59]), indicating that there is no evidence of publication bias or small-study effects. To confirm the findings of Egger's test in a more systemic way, we used a trim-and-fill approach. This approach is a nonparametric (rank-based; not required to fit a normal distribution) data augmentation technique, and it checks publication bias among the calculated ESs and, when necessary, removes studies that cause asymmetry, fills in hypothetical studies to secure symmetry of the

funnel plot, and re-estimates revised ESs (Duval & Tweedie, 2000). Results of this approach showed that no trimming was performed and the data were unchanged, indicating that there is no identifiable sign of publication bias or small-study effects from the calculated ESs.

Coding Scheme

In order to examine whether the intervention effect varied as a function of several intervention and contextual factors, we coded methodological variation in the interventions in the following features. First, intervention features included target language (whether the intervention targeted L1 or L2); teacher training support (whether teacher training was included or not, and if included, whether ongoing support was included or not); intervention domain (whether combined with math or reading only); Information Communication Technology (ICT) inclusion (whether ICT was included or not), such as the use of computers, mobile phones, e-readers, tablets, and audio/video; community involvement (whether community was involved as part of the literacy intervention or not); and intervention duration (either < 1 month, between 1 and 6 months, between 6 months and 1 year, or longer than 1 year). Intervention duration was coded as the time between pretest and posttest. Second, contextual factors included urbanicity (either urban, rural, or other—mixed or not specified) and region (either Europe & Central Asia, South Asia, Middle East & North Africa, South East Asia & Pacific, Sub-Saharan Africa, or Latin America & Caribbean; in accordance with the World Bank's regional country classification).

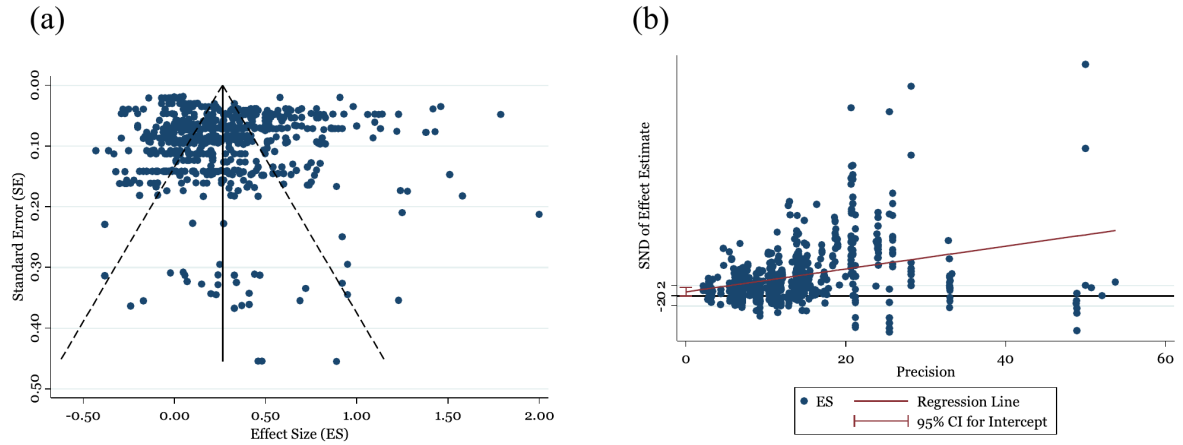


Figure 3. Funnel plot (a) and Egger's test (b) for publication bias. [Color figure can be viewed at wileyonlinelibrary.com]

It should be noted that other key information, such as gender (the number of girls vs. boys), special education (number of children served in special education), disability status, socioeconomic status, material development (whether the development of materials such as textbooks were part of the intervention), and intervention intensity (e.g., hours per session, sessions per week), was identified and coded. However, they were excluded in our analyses due to too much missing data or little variation.

In order to establish a high level of interrater reliability, the first and second authors of this study independently undertook the initial coding for a subset of the studies (25%), resolved any discrepancies through discussion, and reached full agreement before the remaining studies were coded.

Data Analytic Strategies

STATA 14 with *meglm* command was used to conduct a multilevel regression model analysis, and it was replicated using HLM 7 (Raudenbush, Bryk, & Congdon, 2011). Considering that we already have the variance of the observed values of the outcome (i.e., standard errors of the ESs), it makes multilevel modeling for a meta-analysis distinguishable from a regular multilevel model. In other words, our models are Level-1 variance-known models (Raudenbush & Bryk, 2002); and the *meglm* command in STATA 14 as well as HLM 7 support this approach (Lee et al., in press).

To answer the first research question, we computed (a) overall average ES, and (b) average ESs for each outcome (i.e., emergent literacy skills, word and syllable, reading fluency, reading comprehension, and oral language) along with contrasts of average ESs between the outcomes. The overall average ES was obtained in an unconditional model without any independent variable:

$$ES_{ijk} = \gamma_{000} + r_{0jk} + u_{00k}, \qquad (1)$$

where ES_{ijk} is the observed ES i in sample j from study k, γ_{000} is the overall average ES, r_{0jk} is Level-2 variance, and u_{00k} is Level 3-variance. Level-1 variance is omitted in the equation because we already knew and included it in the estimation (i.e., Level-1 variance-known model).

To estimate average ESs for each of the five outcomes as well as whether the ES estimate of an outcome (e.g., emergent literacy skills) is statistically different from that of another outcome (e.g., reading comprehension), a conditional model was fitted where a variable for outcome type categories was included as an independent variable (i.e., a bivariate regression model). For average ESs for each outcome, the following equation has four dummy variables for the outcome types with the emergent literacy skills type the reference (or omitted) category:

$$\begin{aligned} ES_{ijk} = \gamma_{000} &+ \gamma_{100} \times \text{WordSyllable}_{ijk} + \gamma_{200} \times \text{ReadingFluency}_{ijk} \\ &+ \gamma_{300} \times \text{ReadingComprehension}_{ijk} \\ &+ \gamma_{400} \times \text{OralLanguage}_{ijk} + r_{0jk} + u_{00k} \qquad (2) \end{aligned}$$

To answer the second research question, we conducted moderation analyses for both the overall estimate (Model 1) and each language and literacy outcome (Models 2–6) by fitting conditional models (cf. Equation 2) with all 11 moderator variables (i.e., a meta-regression). A series of binary variables were created for the moderator variables. For example, for the teacher training variable which was coded as three categories (no PD, one-time PD only, and PD plus ongoing support), three binary variables were used with one reference category omitted. This allowed us to estimate ES contrasts as a function of the moderator variable (i.e., coefficients of each category for moderator variables), after keeping other variables and their values at their means (i.e., holding others constant). As mentioned earlier, the publication type and participant number variables were controlled for the moderator analyses to minimize possible publication bias or small-study effects.

Results

The distribution of all ES estimates (N = 661) from 129 samples (which are from 67 studies) is shown in Figure 4. The calculated ESs ranged from −.43 to 2.00, with a standard deviation of .34. Approximately 80% of the ESs (n = 532) were positive, whereas 20% (n = 129) had negative ESs from 45 samples (34.9%). Figure 4 shows that most of the ESs (approximately 54.6%) were clustered around between 0 and .50, whereas approximately 20.9% (n = 138), and 7.6% (n = 50) were medium ESs ($.5 \leq d < .8$) and large ESs ($d > .8$), respectively.

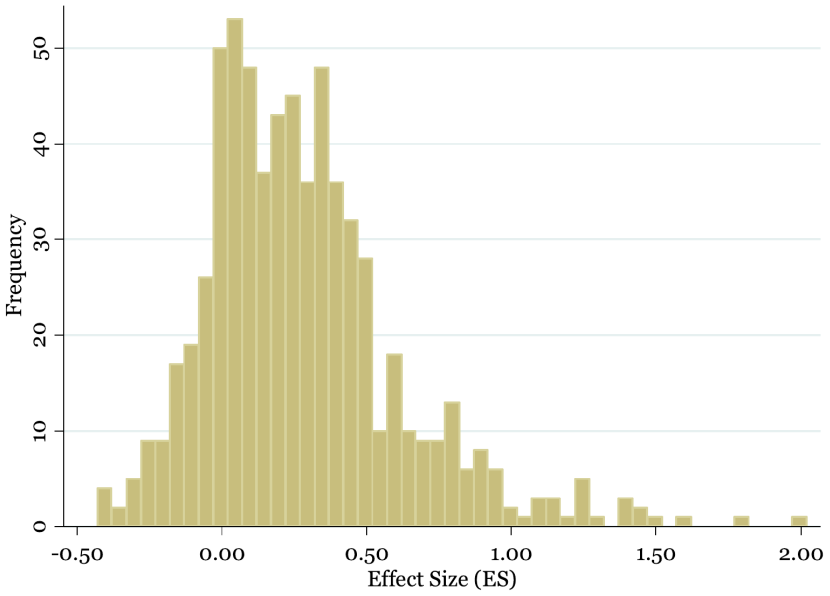


Figure 4. Histogram of effect size estimates. [Color figure can be viewed at wileyonlinelibrary.com]

RQ 1: What Is the Impact of Multicomponent Literacy Interventions in LMICs on Language and Reading Outcomes?

Table 1 presents the results of mean ES estimates overall and for each outcome. The overall average ES was .30 ($p < .001$; based on 661 ESs from 67 studies representing 128 samples), indicating that multicomponent literacy interventions in LMICs have an overall positive impact on literacy outcomes. Furthermore, we examined the ESs for each outcome based on 623 ESs from 113 samples (which are from 61 studies). A total of 38 ESs from 19 studies representing 34 unique samples were excluded in Model 2 (see Table 1) because these ESs could not be categorized into one of the five literacy outcomes types. For example, they included composite scores, standardized test scores, or percentages of successful readers. Results were as follows: emergent literacy skills (.40), word reading (.32), reading fluency (.28), reading comprehension (.25), and oral language (.20). There were statistically significant differences across the outcomes such that the ES for emergent literacy skills was larger than those for the other outcome types, such as word reading ($p < .01$), reading fluency ($p < .001$), reading comprehension ($p < .001$), and oral language ($p < .001$). In addition, ES for oral language was smaller than word reading ($p < .001$).

Table 1
Mean Effect Size (ES) Estimates

Models	No. of ESs	No. of samples	No. of studies	Mean ES	*SE*	Contrasts ($p < .05$)
Model 1: Overall (unconditional model)	661	129	67	.30***	(.03)	—
Model 2: By outcome type (conditional model)						
A. Emergent literacy skills	151	79	47	.40***	(.04)	A > B, C, D, E
B. Word & syllable	159	73	44	.32***	(.04)	B < A; B > D, E
C. Reading fluency	116	78	43	.28***	(.04)	C < A; C > E
D. Reading comprehension	94	71	45	.25***	(.04)	D < A, B
E. Oral language	103	65	34	.20***	(.04)	E < A, B, C

Note. Reported contrast results are statistically significant at 5% or lower levels.
***$p < .001$.

RQ 2: Do the Effects Vary as a Function of Intervention Features as Well as Contextual and Other Factors?

Table 2 describes average ESs for moderator variables along with contrasts results between values for each moderator coming from six independent meta-regression models, where Model 1 is for overall outcomes (661 ESs, 129 samples, and 67 studies), Model 2 for emergent literacy skills (151 ESs, 79 samples, and 47 studies), Model 3 for word reading (159

ESs, 73 samples, and 44 studies), Model 4 for reading fluency (116 ESs, 78 samples, and 43 studies), Model 5 for reading comprehension (94 ESs, 71 samples, and 45 studies), and Model 6 for oral language (103 ESs, 65 samples, and 34 studies).

Model 1: Overall

The results suggest that the effects of literacy intervention across the outcomes varied as a function of teacher training support, such that interventions with PD with ongoing support (d = .32, p < .001) had a larger average effect than those with one-off PD (d = .11, p > .05; ES difference = .21, p < .05) and no PD (d = .01, p > .05; ES difference = .30, p < .05). However, there was no statistically significant difference between the no PD and one-off PD conditions (d = .11, p > .05; ES difference = .09, p > .05; Note that some of the values [e.g., the difference between .11 and .01 is not .10] do not add up due to rounding.).

Model 2: Emergent Literacy Skills

The larger effect of providing PD and ongoing support (d = .40, p < .001) than providing no PD (d = .03, p > .05) was also found in the emergent literacy skills (ES difference = .37, p < .05). Interventions that included ICT (d = .58, p < .001) were more effective than those that did not (d = .29, p < .001; ES difference = .30, p < .05). In contrast, the involvement of the community in the literacy interventions was associated with a smaller mean ES (d = .25, p < .001) than ones with no community involvement (d = .44, p < .001; ES difference = .19, p < .05). Furthermore, the mean intervention effect was larger in the Latin America and Caribbean region (d = .70, p < .001) than in the South East Asia and Pacific region (d = .29, p < .05; ES difference = .41, p < .05). Although ES differences in the intervention domain and intervention duration were ≥ .20, they were not statistically significant (p > .05).

Table 2

Average Effect Sizes (ES) for Moderator Variables From Meta-Regression Analysis

Moderator variable	Model 1: Overall 661 ESs 129 Samples 67 Studies			Model 2: Emergent literacy skills 151 ESs 79 Samples 47 Studies			Model 3: Word and syllable 159 ESs 73 Samples 44 Studies			Model 4: Reading fluency 116 ESs 78 Samples 43 Studies			Model 5: Reading comprehension 94 ESs 71 Samples 45 Studies			Model 6: Oral language 103 ESs 65 Samples 34 Studies		
1. Target language																		
(1) First language	.31***	(.03)	No diff.	.40***	(.04)	No diff.	.31***	(.04)	No diff.	.24***	(.03)	No diff.	.24***	(.04)	No diff.	.27***	(.04)	(1) > (2)
(2) Second language	.24***	(.06)		.28**	(.10)		.25*	(.11)		.20**	(.08)		.25*	(.10)		.09	(.06)	
2. Teacher training support																		
(1) No PD/taught by researchers	.01	(.12)	(3) > (2), (1)	.03	(.18)	(3) > (1)	−.18	(.30)	No diff.	−.14	(.23)	No diff.	−.47*	(.20)	(3), (2) > (1)	−.02	(.24)	No diff.
(2) One-time PD only	.11	(.10)		.22	(.13)		.22	(.16)		.13	(.13)		.24*	(.11)		.09	(.11)	
(3) Ongoing support	.32***	(.03)		.40***	(.04)		.32***	(.04)		.25***	(.03)		.25***	(.04)		.23***	(.04)	
3. Intervention type																		
(1) Reading only	.31***	(.03)	No diff.	.42***	(.05)	No diff.	.32***	(.04)	No diff.	.26***	(.03)	No diff.	.28***	(.05)	No diff.	.22***	(.04)	No diff.
(2) Combined with Math	.22*	(.09)		.18	(.12)		.22	(.11)		.09	(.09)		.07	(.14)		.15	(.08)	
4. ICT use																		
(1) Yes	.30***	(.07)	No diff.	.58***	(.11)	(1) > (2)	.30***	(.09)	No diff.	.22*	(.10)	No diff.	.19	(.12)	No diff.	.16**	(.05)	No diff.
(2) No	.29***	(.03)		.29***	(.05)		.30***	(.04)		.24***	(.03)		.25***	(.05)		.27***	(.06)	
5. Community involvement																		
(1) Yes	.22***	(.05)	No diff.	.25***	(.07)	(2) > (1)	.19**	(.07)	No diff.	.18***	(.04)	(2) > (1)	.19**	(.07)	No diff.	.21***	(.06)	No diff.
(2) No	.34***	(.04)		.44***	(.05)		.36***	(.04)		.33***	(.05)		.28***	(.05)		.20***	(.04)	
6. Intervention length																		
(1) ~1 month	.14	(.17)	No diff.	−.04	(.36)	No diff.	.15	(.29)	No diff.							.06	(.22)	(2) > (1), (3)
(2) 1–6 months	.33***	(.08)		.58***	(.14)		.26**	(.09)		.14	(.10)	No diff.	.12	(.11)	No diff.	.49**	(.16)	
(3) 0.5–1 year	.27***	(.04)		.34***	(.06)		.26***	(.06)		.25***	(.05)		.24***	(.06)		−.01	(.11)	
(4) 1 year–	.31***	(.05)		.40***	(.07)		.37***	(.06)		.25***	(.05)		.31***	(.06)		.09	(.11)	
7. Urbanicity																		
(1) Urban	.18	(.10)	No diff.	.35*	(.14)	No diff.	.26	(.15)	No diff.	.32**	(.11)	No diff.	.32*	(.15)	No diff.	.03	(.14)	No diff.
(2) Rural	.34***	(.05)		.40***	(.07)		.29***	(.06)		.22***	(.06)		.25***	(.07)		.33***	(.07)	
(3) Other (e.g., mixed)	.29***	(.04)		.37***	(.05)		.32***	(.05)		.23***	(.04)		.21***	(.06)		.21***	(.06)	
8. Region																		
(1) Sub-Saharan Africa	.27***	(.04)	No diff.	.40***	(.05)	(6) > (4)	.31***	(.04)	(3) > (6)	.27***	(.04)	(3) > (4), (6); (3), (5), (1) > (6)	.26***	(.06)	No diff.	.26***	(.07)	No diff.
(2) Europe & Central Asia	.24	(.19)		−.03	(.40)								.21	(.37)		.00	(.17)	
(3) Middle East & North Africa	.56**	(.18)		.53	(.33)		.58**	(.21)		.53**	(.18)		.53	(.27)		.47	(.26)	
(4) South East Asia & Pacific	.30***	(.08)		.29**	(.10)		.31**	(.11)		.10	(.09)		.20*	(.10)		.24	(.14)	
(5) South Asia	.36***	(.07)		.26*	(.10)		.28**	(.10)		.29***	(.07)		.20*	(.10)		.41***	(.12)	
(6) Latin America & Caribbean	.33**	(.11)		.70***	(.16)		−.02	(.16)		−.03	(.12)		.14	(.16)		.16	(.13)	

Note. Values are average ESs for each category of moderator variables computed from meta-regression and standard errors are in parentheses. Third columns for each model include contrasts between values for each category at the 5% significance level. Empty cells indicate that there were not enough values to compute relevant statistics. Types of publications and numbers of participants were controlled. PD = professional development; ICT = information and communications technology. *p < .05, **p < .01, ***p < .001.

Model 3: Word Reading

Average impact of the literacy intervention was larger in Middle East and North Africa (d = .58, p < .01) than in Latin America and Caribbean (d = −.02, p > .05; ES difference = .60, p < .05). For teacher training support, although large mean ES differences were found (d = .32, p < .001 for PD with ongoing support; d = −.18, p > .05 for no PD), they were not statistically significant (p > .05) due to their relatively large standard errors.

Model 4: Reading Fluency

Mean ES was smaller when the intervention included community involvement (d = .33, p < .001 for no community involvement; d = .18, p < .001 for with community involvement; ES difference = .15, p < .05). When it comes to regions, the mean ES was larger in Middle East and North Africa (d = .53, p < .01) than in the South East Asia and Pacific region (d = .10, p > .05; ES difference = .44, p < .05) and Latin America and the Caribbean (d = −.03, p > .05; ES difference = .56, p < .05). In addition, the mean intervention effect was smaller in the Latin America and the Caribbean (d = −.03, p > .05) than in Middle East and North Africa (as shown above), the South Asia (d = .29, p < .001; ES difference = .32, p < .05), and the Sub-Saharan Africa (d = .27, p < .001; ES difference = .30, p < .05).

Model 5: Reading Comprehension

For the reading comprehension outcome, we found that receiving one-time PD (d = .24, p < .05) and receiving PD and ongoing support (d = .25, p < .001) both had larger impact than no teacher training support (d = −.47, p < .05; ES differences = .71, p < .01; .72, p < .001; respectively). In other words, on average, interventions that had one-off teacher training and those with PD and ongoing support had larger effects compared to those without any teacher training.

Model 6: Oral Language

Using first language as a medium of the literacy interventions (d = .27, p < .001) was more effective than using second language for improving children's oral language in LMICs (d = .09, p > .05; ES difference = .17, p < .05: the ES difference is not .18 due to rounding.). In addition, interventions lasting between 1 and 6 months (d = .49, p < .01) were more effective than those conducted for < 1 month (d = .06, p > .05; ES difference = .43, p < .05) and between 6 months and 1 year (d = −.01, p > .05; ES difference = .50, p < .05).

Discussion

In this study, we investigated the effect of literacy instruction on reading and language achievement for children in LMICs. Despite the impact of reading development on individuals' income, health, and empowerment, as well as the economic development of a country, in many LMICs, reading has not been taught explicitly as a skill and the majority of children cannot read a single word even after years of instruction (Gove & Cvelich, 2011). In the last decade, however, considerable attention and resources have been invested in an effort to address this challenge in LMICs. In this study, we examined the impact of this effort by conducting a meta-analysis of the available evidence.

Overall, the results from 213,464 participants in 67 studies from 32 countries have revealed positive average effects of literacy interventions in LMICs across various contexts. The mean ES across the studies and across the outcomes was .30. This average ES is quite impressive and promising, considering the many educational challenges in LMICs described above and given that in educational interventions, ESs of approximately .20 are considered substantial (Hill, Bloom, Black, & Lipsey, 2008). The overall average ES of .30 is also in line with a recent meta-analysis on student learning in Sub-Saharan Africa (Conn, 2017) despite the differences in the outcomes and scope between Conn's (2017) study and the present meta-analysis. These findings indicate that concerted efforts in explicit and systematic instruction in reading in resource lean environments result in substantial improvement in children's reading and language outcomes.

A closer look revealed large variation in effects, ranging from negative effects to large positive effects. Specifically, negative results were reported in approximately 20% of the ESs. These results indicate challenges in making a positive impact on children's reading skills in the LMIC context. As noted above, challenges abound in multiple aspects in this

context. This was apparent in several reports which documented difficulties, including large class size (e.g., over 300 children in a single cramped classroom taught by two teachers, Dowd & Mabeti, 2011), absenteeism (e.g., Dowd & Mabeti, 2011, Friedlander et al. 2012), implementation and fidelity (e.g., not being able to complete teacher training, teachers reverting to old practices, and books not being used as intended for fear of loss or damage, AIR, 2014; RTI, 2009) to name a few. Therefore, the positive impact found in the majority of studies, despite these challenges, is promising.

Beyond the overall effect, of particular interest is the variation in average effects across different reading and reading-related outcomes, with the largest average effect for emergent literacy skills (.40), followed by word reading (.32), reading fluency (.28), reading comprehension (.25), and oral language (.20). The differential effects of interventions on various reading and language outcomes were also noted in recent reviews of studies in LMICs (Conn, 2017; Graham & Kelly, 2018; Kim et al., 2016). Together, these findings indicate that with explicit and systematic instruction, lower level skills, such as emergent literacy skills, are relatively easier to improve than higher order skills such as reading comprehension. Emergent literacy skills such as phonological awareness and knowledge of orthographic symbols (letter name and sound knowledge) are essential to lexical-level literacy skills such as word reading and spelling development (National Early Literacy Panel, 2008; National Institute of Child Health and Human Development, 2000), and considered constrained skills because the amount of learning in these areas is finite (Snow & Matthews, 2016). In contrast, reading comprehension is a higher order skill that requires all the other skills (emergent literacy skills, word reading, reading fluency, oral language) as well as higher order cognitive skills such as inference, perspective taking, and comprehension monitoring; Barnes, Ahmed, Barth, & Francis, 2015; Cain, Oakhill, & Bryant, 2004; Cromley & Azevedo, 2007; Kim, 2017; Oakhill, Cain, & Bryant, 2003; Perfetti, Landi, & Oakhill, 2005). Therefore, improving reading comprehension requires concerted and sustained efforts to a greater extent compared to constrained skills such as emergent literacy skills. Furthermore, oral language is considered an unconstrained skill (or a large problem space, Snow & Kim, 2007) as it is acquired gradually through long-term experience (Paris & Paris, 2003; Snow & Matthews, 2016).

Another important research question was differential effects—or lack thereof—as a function of various intervention features, including whether the intervention targeted students' L1 versus L2, whether teacher training included one-off PD or PD plus ongoing support, whether the intervention focused on literacy only or also included math, whether the intervention included ICT or community involvement, and how long the intervention lasted. Overall results were not consistent across outcomes. One moderator that had a relatively consistent result was teacher training support—the average effect was larger when teachers were provided with ongoing PD beyond a one-off workshop or no training. This is consistent with literature on teacher PD (Akyeampong et al., 2013; Bunyi, Wangia, Magoma, Limboro, & Akyeampong, 2011; Dubeck, Jukes, Brooker, Drake, & Inyega, 2015; Education Development Center, 2014a, 2014b; Piper & Korda, 2011; Pouezevara et al., 2013), and indicates the importance of continued support for teachers in implementing newly acquired instructional practices. Given that many teachers in LMICs do not have previous experience of teaching reading, thus, lack foundational knowledge about reading development and instruction, this is particularly crucial in these countries (see Kim et al., 2016 for a narrative review).

Larger ESs were also found for interventions that targeted L1 than in L2 in oral language. These larger effects in L1 instruction are consistent with the idea that learning is facilitated when input is comprehensible to children; one of the essential components for comprehensibility is the language of instruction (Ball, 2011). Beyond the importance of comprehensible input, these findings are notable in relation to the benefit of literacy instruction in L1 for literacy acquisition in L2, known as cross-language transfer (Cummins, 1979, 1984; Piper, Zuilkowski, & Ong'ele, 2016; Vaughn et al., 2006; Wawire & Kim, 2018). It is notable, though, that the larger effect for L1 instruction was found only in oral language but not in literacy outcomes. This might indicate that with explicit and systematic instruction, effects on literacy outcomes may be as effective when instruction is in students' L1 or L2. However, this speculation requires further investigation.

We also found a larger ES with ICT use, such that the mean ES was larger for the emergent literacy outcomes, indicating the potential promise of ICT in improving children's literacy skills in low- and middle-income country contexts. For example, in literacy intervention conducted in Kenya, tablet PCs were used by trained tutors to upload classroom observations and to support instructional coaches (e.g., monitoring and changing their coaching behavior) as well as to improve instructional competencies of the teachers (Piper, Zuilkowski, Kwayumba, & Strigel, 2016; RTI International, 2015a). In other interventions, prerecorded interactive audio instruction was included as part of teacher support materials to help local teachers improve their instructional techniques and strategies specific to reading in their classrooms (Education Development Center, 2014a, 2014b, 2015).

A somewhat surprising finding was the lack of clear differential effects as a function of intervention duration. We speculated that interventions with longer duration would result in greater effects in reading outcomes, particularly for unconstrained skills noted above such as oral language or higher order skill such as reading comprehension. However, our findings showed that, on average, intervention duration did not make a difference in the ESs in the majority of outcomes after accounting for the differences in the other factors examined in this study (see results in Table 2). An exception was in oral language where interventions between 1 and 6 months had larger average effects than those with other durations. Explanations for these results— why the effects are larger for the particular time span examined in oral language, and the lack of differences in reading outcomes—are not clear. However, one potential reason may include large variation in the nature of studies of the included studies. For example, a study in Zambia examined the effect of a specific computer-based phonics game, GraphoGame, on first grade students' emergent literacy skill (orthographic awareness), spelling, and vocabulary skills. Students in the treatment condition were exposed to the phonics game for upto 190 min within 2 week duration (Jere-Folotiya et al., 2014). In contrast, other studies implemented a comprehensive literacy program targeting multiple components necessary for reading development. For instance, the Literacy Boost program is a multicomponent program and was implemented in Bangladesh for 2 years (Jonason et al., 2014). Similarly diverse set of studies were included for oral language outcomes. For example, Opel, Ameer, and Aboud (2009) as well as Ergül et al. (2016) implemented dialogic reading for 4 weeks and 7 weeks, respectively, and found an effect on children's vocabulary. On the other hand, other studies targeted multiple skills including oral language as part of a comprehensive reading program for longer periods (e.g., 10 months or longer; e.g., Guajardo et al., 2012; Piper & Kwayumba, 2014). If a study targets a very specific skill (e.g., an emergent literacy skill such as orthographic awareness) that requires a relative short duration and finds an effect on the target skill, and another study provides a comprehensive intervention that targets overall reading development (i.e., reading outcomes at multiple levels—word and discourse) and finds effects on multiple outcomes, then, intervention duration would not yield differences in ESs. Thus, lack of significant differences in ESs as a function of intervention duration may be due to differences in the nature of studies. However, it is also possible that the results are due to systematic differences among the studies beyond what was captured in the intervention and contextual factors accounted for in this study. Carefully designed randomized control trials are needed to shed light on the effects of intervention duration and intensity on ESs.

Another counterintuitive finding was a smaller effect of interventions that included community involvement than those that did not on emergent literacy skills and reading fluency. Evidence from high-income countries as well as LMICs has consistently identified relations between language and literacy activities in the home and children's literacy acquisition (Bus et al., 1995; Chansa-Kabali et al., 2014; Dowd & Friedlander, 2016; Dynia et al., 2014). However, the evidence has been largely correlational, particularly in LMICs. The current meta-analysis revealed that the effect of multicomponent literacy interventions that incorporate community component may not yield a larger effect than interventions that focus on instruction in the school. However, this finding should be interpreted with caution because quality of implementation and other factors have not been taken into consideration. The nature of community involvement as well as quality indicators of community involvement (community buy in; fidelity of implementation) are important factors for results. For instance, Banerjee et al. (2010) compared two types of community involvement—(a) training community members on reading assessment so that they know their children's reading status and (b) training on reading instruction strategies—and found no effect of the training on assessment but a positive effect of the training on reading instruction. Furthermore, several studies indicated substantial challenges in parents' and/or students' participation in community sessions such as parent

workshops and student reading camps (e.g., Badiable, Guajardo, Fermin, & Robis, 2013; Brown, 2013). Of course, without quality programming and implementation, community involvement would not translate to the anticipated positive outcomes on children's reading skills.

To our knowledge, there is only one randomized control study in a LMIC that directly examined the additive effect of community involvement on reading outcomes by holding other things constant —comparing the effects of a teacher training only condition with a teacher training plus community involvement condition (including workshops on how to improve reading in the community setting). Conducted in Rwanda, this study found a positive effect of community involvement compared to the teacher training only condition (ES differences ranging from .12 to .29; Friedlander & Goldenberg, 2016). Future studies should further document challenges in implementation and associated solutions, and investigate fidelity of implementation and mechanisms (e.g., whether attendance in workshops and other activities translate to changed behaviors in the home and community, which, then, result in improvements in children's reading and other outcomes) to unpack the nature and true effect of community involvement on student outcomes.

When it comes to the effects of contextual factors, some differences in mean ESs were found as a function of region. Although there was no difference in mean ESs in overall outcomes, some differences were found in specific literacy outcomes such that mean ES in Latin America and Caribbean region was larger than that in South East Asia and Pacific region in the emergent literacy outcome, whereas it was smaller in word reading and reading fluency, after accounting for the intervention features and the other contextual factors included in this study. Explanations for these differential effects remain unclear, particularly given the inconsistent effects across different outcomes. Note, however, that caution should be exercised due to the small number of studies (i.e., 44 ESs from 5 studies) in Latin America and the Caribbean; future studies are needed to shed light on factors that influence variation in ESs in different regions and inform future efforts in improving children's reading skills across regions.

Finally, there were no statistically significant differences between rural and urban areas, with the exception of the oral language outcome, where ESs were larger in rural and other areas than in urban areas. Typically, rural areas have more limited resources than urban areas, such as teacher quality (e.g., McEwan, 1999). However, resources in urban areas also tend to vary widely in different contexts (e.g., nonformal settlement areas vs. relatively affluent areas). Overall, these findings may indicate that explicit and systematic instruction on reading skills with teacher training can yield similar effects in the majority of outcomes regardless of urbanicity.

Limitations, Future Directions, and Implications

There are several limitations that should be considered for interpreting the findings in this study. First, generalizability of the findings is limited to contexts similar to those examined in this study—LMICs with limited resources. Second, one critical factor in estimating the effects of multicomponent intervention is implementation quality—fidelity. Unfortunately, this information was not available in many of the studies and thus, not included in the present analysis.

The present meta-analysis suggests several areas of future direction for research and practice. First, future efforts are needed to shed light on instructional approaches to improve advanced, unconstrained skills such as reading comprehension, oral language, and higher order cognitive skills. Evidence clearly indicates that oral language skills are essential to reading comprehension development. However, oral language takes a prolonged time to develop and evidence-based instructional approaches to improve oral language in LMICs, where large classes are common, are limited. This is compounded by the lack of training on teaching reading for meaning that prevails in many LMICs, which may reduce children's opportunity to practice their oral language skills (Akyeampong et al., 2013; Vavrus & Bartlett, 2012). Identifying effective approaches to oral language instruction is particularly important in LMICs where students are learning to read in L2; and therefore, reading proficiency in L2 depends on oral language proficiency in L2. Furthermore, reading comprehension and listening comprehension require development of higher order cognitive skills such as inference, perspective taking, and comprehension monitoring (Cain et al., 2004; Florit et al., 2014; Kim, 2016, 2017; Lepola, Lynch, Laakkonen, Silvén, & Niemi, 2012; Tompkins, Guo, & Justice, 2013). However, unclear from the studies in the present meta-analysis is the extent to which these were explicitly addressed in reading instruction.

Another important future direction is further systematic examination of factors that maximize students' literacy acquisition in LMICs, including instructional intensity (how many minutes per day and how many sessions per week). Our attempt to address the question about instructional intensity was constrained by the limited information provided in the reviewed studies (i.e., many studies did not indicate minutes of instruction per day or frequency of instruction per week), and thus, we used the duration of the studies (i.e., time between pretest and posttest) as a proxy. The effect of community involvement also requires further investigation, particularly in LMICs, to elucidate the feasibility and mechanisms of relations as well as its causal impact.

Moreover, all these questions should be also addressed for subgroups of students (e.g., those with learning disabilities, visual and physical disabilities). Interventions in LMICs so far have principally focused on developing reading curriculum, associated materials, and PD. Although this is necessary and large strides have been made in these areas, the next step should consider promoting literacy instruction for all children with various needs. This is in line with the UN's Inclusive and Quality Education For All (see the Sustainable Development Goal 4). The challenges for differentiating instruction in LMICs are many, large class size being a major one, and should be addressed in future efforts. Finally, the present review focused only on reading and reading-related outcomes, but not writing. This reflects the reality of where the field is— not a single study in our review examined writing outcomes beyond spelling. Given that writing is also an essential skill in modern societies, and it interactively develops with reading skills, future efforts should include integrating writing instruction with reading.

Findings from this study offer several important practical implications. First, the positive overall effects of multicomponent reading instruction across different contexts in the world provide strong evidence for adopting the policy of explicit and systematic reading instruction. Reading skills have consequences for individual development as well as for society at large. Second, successful implementation of explicit reading instruction requires teacher training. Our findings were consistent with the literature, indicating a need for teacher PD that consists of intensive workshops followed by booster training sessions and ongoing support. This is the case not only for preservice teacher training but also for in-service teacher training—the latter is critical particularly in LMICs because the extant teacher workforce has generally received little training on reading instruction. Third, language of instruction is a hotly debated policy issue around the globe. Literature from developmental and cognitive science is clear about the advantage of learning in L1 not only in content learning but also for literacy acquisition in L2 (see literature on cross-language transfer; Vaughn et al., 2006; Wawire & Kim, 2018). However, policy on the language of instruction in many multilingual contexts is complex, due to several reasons including the perceived higher status of official languages (L2; Trudell & Piper, 2014), the lack of qualified teachers who can teach in children's L1 or materials to use in teaching, and the challenges of multilingual settings where children have many different L1s (see Kim et al., 2016 for a review). Although these details are beyond the scope of this study, our findings do indicate that intervention effects on students' oral language skills are larger when the intervention is in L1, though students can benefit in learning literacy skills in either L1 or L2 when instruction is explicit and systematic.

In this study, we have conducted a comprehensive and systematic review of the studies using meta-analysis to examine the causal impact of multicomponent literacy instruction in LMICs. Recent efforts in the area of reading in LMICs are highly encouraging and should be continued and supported. Future investigations on their long-term effects and effects beyond reading outcomes (e.g., graduation rate, drop-out rate, career attainment) are warranted.

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